ContractLens MAS — Comparative Synthesis Report

Unforeseen Physical Conditions — Remnant Piles

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Agents: GCC Agent · NEC Agent · Comparison Agent
Workflow: Query Decomposition → Parallel Querying → Cross-examination → Synthesis → QA
Overall verdict: **PASSED**

Context / Scenario

Drainage improvement project in Tuen Mun. Excavation reveals previously unrecorded remnant piles. Works are suspended for three weeks while the Engineer/PM instructs redesign. The contractor seeks additional cost (HK$1.8M) and extension of time.

Topic

Unforeseen physical conditions: a contractor on a Hong Kong drainage project encounters remnant piles not indicated on the contract drawings, causing a three-week delay and approximately HK$1.8 million of additional cost. Compare the entitlement, notification procedure, and assessment approach under the relevant contract.

Questions

Entitlement, notification, and assessment for unforeseen physical conditions.

Query Plan

L1 — Spirit / Philosophy: How does the contract's governance philosophy allocate the risk of unforeseen physical conditions and encourage timely, cooperative management of ground-condition surprises?
L2 — Clause-level Provisions: What clauses establish entitlement, notice requirements, time and cost relief, and the method for assessing delay and additional cost where remnant piles or other unexpected physical obstructions are encountered?
L3 — Practice Implications: In practice, what behaviours should the contractor, contract administrator, and project team adopt to preserve entitlement, evidence causation, mitigate delay, and achieve a fair assessment?
Canonical topic: physical_conditions

Anchor Clauses

GCC: Clause 13, Clause 14, Clause 50(1), Clause 50(1)(a), Clause 64
NEC4: cl. 60.1(12), cl. 15.1, cl. 15.2, cl. 63.1, cl. 63.5

Comparative Synthesis

> **Executive summary —** GCC generally bars automatic relief for remnant piles unless an Engineer's written variation unlocks time/cost claims; NEC grants a compensation event if cl. 60.1(12) is met. GCC is Engineer-led and retrospective; NEC is collaborative and prospective, requiring earlier warning, quotation and programme evidence.

1. Spirit / Philosophy — Risk transfer versus relational governance.

GCC: The standard GCC allocates unexpected physical-condition risk primarily to the Contractor through deemed site inspection, no-payment, and all-risks tender provisions. Its governance is Engineer-led and claims-oriented: relief is not automatic, but depends on instruction, written variation, notices, particulars,

records, and later assessment. In Transaction Cost Economics terms, it emphasises ex ante risk allocation and formal control to reduce opportunistic claims.

NEC: NEC4 ECC treats a qualifying unforeseen physical condition as a compensation event managed through early warning, mitigation, quotation, and implementation. The regime is grounded in mutual trust and co-operation under cl. 10.2, using structured deadlines to force timely information exchange. TCE-wise, NEC relies on relational governance and adaptive processes to reduce the cost of late dispute resolution and maladaptation.

2. GCC Position (Implementation) — Relief depends on Engineer-led variation.

Implementation flow:

1. Contractor discovers remnant piles and notifies Engineer/request instruction if design cannot proceed (Clause 13 / Clause 15) →
2. Engineer may issue written variation to overcome obstruction (Clause 60(1); Clause 61 / Clause 61(1)) →
3. Contractor gives delay notice within 28 days of instruction/event (Clause 50(1)(a)) →
4. Contractor gives cost notice within 28 days after event (Clause 64(2)) →
5. Contractor mitigates, proceeds, records and particularises (Clause 86(1); Clause 64(3); Clause 50(3); Clause 64(4); Clause 64(6)) →
6. Engineer assesses EOT/cost and programme is revised if EOT granted (Clause 50(2); Clause 63; Clause 50(4)).

Under the GCC, the corpus does not use “Project Manager”, “Prices”, or “compensation events”; administration is by the Engineer, time runs against the Time for completion, and payment feeds the Final Contract Sum based on Bills of Quantities. Clause 13(1) deems the Contractor to have inspected the Site. Clause 13(2) bars additional payment for incorrect or insufficient Employer information and does not relieve the Contractor from risk or obligation “on the ground that he did not or could not foresee any matter which may in fact affect or have affected the execution of the Works.” Clause 14 deems Tender rates to cover all risks, liabilities and obligations. Thus remnant piles do not automatically give time or money.

If the obstruction makes the original design physically impossible, the Contractor must seek the Engineer’s instruction under Clause 15. The Engineer may order a written variation under Clause 60(1), such as changed drainage alignment or depth. Only that formal variation unlocks potential relief; valuation is under Clause 61 / Clause 61(1), using Contract rates under Rules 1 and 2 or a fair rate under Rule 3. Clause 50(1)(b)(iv) was cited in the GCC clause set for EOT entitlement.

Notice is strict. For time, the Contractor must give written notice of the cause and probable extent of delay within 28 days of the relevant instruction or event under Clause 50(1)(a). The Engineer determines EOT taking account of all circumstances known to him under Clause 50(2), usually by critical-path and float analysis. If detailed particulars are missing, the Engineer assesses only to the extent possible on available information under Clause 50(3). Whenever EOT is granted, the Contractor revises the programme under Clause 50(4).

For money, Clause 64(2) requires written notice of intention to claim within 28 days after the event. Failure means “such claim shall not be considered” under Clause 64(5). The Contractor must keep “contemporary records as may reasonably be necessary to support any claim” under Clause 64(3), submit a first interim account explaining the sum and calculation under Clause 64(4), and provide particulars; otherwise Clause 64(6) limits assessment to available information. Clause 63 covers actual additional prolongation/disruption costs incurred, excluding profit, not simple extension of Preliminary BQ rates. Clause 86(1) requires the Contractor to proceed with due diligence even during dispute, alongside the mitigation duty.

3. NEC4 Position (Implementation) — CE entitlement with prospective assessment.

Implementation flow:

1. Contractor gives early warning as soon as aware and attends meeting (cl. 15.1; cl. 15.2; cl. 15.3) →
2. Contractor notifies compensation event within eight weeks of becoming aware (cl. 61.3) →
3. Project Manager replies within one week and, if valid, instructs quotation (cl. 61.4) →
4. Contractor submits quotation within three weeks with Prices, Completion Date and Accepted Programme effects (cl. 62.2; cl. 62.3) →
5. Project Manager replies within two weeks: accept, seek revision, or assess (cl. 62.3; cl. 64.1) →
6. Project Manager implements, changing Prices and Completion Date (cl. 66.1; cl. 66.2).

NEC4 ECC handles remnant piles through a prospective, collaborative risk-management regime founded on mutual trust and co-operation under cl. 10.2. cl. 11.2(1), cl. 13.3 and cl. 31.2 were cited in the NEC clause set; the operative response focuses on early warning, compensation events, quotations, Defined Cost and Accepted Programme assessment.

Entitlement arises under cl. 60.1(12) if the physical conditions are within the Site, are not weather conditions, and “an experienced contractor would have judged at the Contract Date to have such a small chance of occurring that it would have been unreasonable to have allowed for them.” cl. 60.2 assumes the Contractor took into account the Site Information, publicly available information referred to in it, information obtainable from visual inspection, and other information an experienced contractor could reasonably be expected to have or obtain. Only the difference between the encountered conditions and those reasonably allowed for is assessed under cl. 60.1(12). If Site Information is inconsistent, cl. 60.3 assumes the Contractor allowed for the more favourable physical condition.

The Contractor should give early warning as soon as aware that the obstruction could increase the total of the Prices or delay Completion under cl. 15.1, and must cooperate with proposals to avoid or reduce effects at early warning meetings under cl. 15.2 and cl. 15.3. The Contractor must continue to Provide the Works and obey PM instructions under cl. 27.3.

The compensation-event notice must be given to the Project Manager within eight weeks of becoming aware under cl. 61.3, or the Contractor loses entitlement to changes in the Prices or Completion Date. The PM replies within one week under cl. 61.4, deciding validity and, if valid, instructing quotation. Under cl. 62.2 and cl. 62.3, the Contractor submits within three weeks a quotation with proposed changes to the Prices, delay to the Completion Date, details of assessment, alterations to the Accepted Programme, the requested three-week delay, and the HK$1.8 million Defined Cost plus Fee change. The PM replies within two weeks under cl. 62.3 and cl. 64.1 by accepting, instructing revision, or making an assessment. Implementation under cl. 66.1 and cl. 66.2 changes Prices and Completion Date.

Assessment is prospective under cl. 63.1: actual Defined Cost before the dividing date, forecast Defined Cost after it, plus Fee. Delay is the movement of planned Completion against the Accepted Programme current at the dividing date under cl. 63.5. Risk allowances for significant chances are included under cl. 63.8. cl. 63.9 assumes competent, prompt reaction and reasonably incurred time and Defined Cost. If early warning was not given, cl. 61.5 and cl. 63.7 assess as if it had been given. Under Options C, D, E and F, cl. 52.2 requires detailed accounts and records of Defined Cost payments.

4. Behavioural Implications — Claims proof shifts to early risk control.

Under GCC, the Contractor’s first behavioural priority is entitlement preservation. On discovering remnant piles, it should notify the Engineer, request an instruction if the design cannot be built, obtain a written variation, and then serve separate 28-day time and cost notices. Evidence is retrospective and defensive: contemporary labour, plant, obstruction, redesign, disruption and time-related cost records must support Clause 64 claims and Clause 50 delay particulars. The Engineer remains the assessment gatekeeper, determining EOT from known

circumstances and valuing variations or actual prolongation cost.

Under NEC, the Contractor's priority shifts to early transparency and forecast management. It must warn the Project Manager as soon as aware, attend early warning meetings, propose mitigation, notify the compensation event within eight weeks, and price the consequence within three weeks of instruction. Evidence is not merely historic proof; it is also forward-looking Defined Cost, Fee, risk allowance and Accepted Programme impact. The Project Manager must respond within one week to CE notice and within two weeks to the quotation, so administration is time-disciplined.

The incentives differ sharply. GCC late cost notice can bar the claim entirely; inadequate particulars narrow what the Engineer can assess. NEC late CE notice bars changes to Prices and Completion Date, while late early warning reduces recoverable consequences and may affect Disallowed Cost under HK guidance for Options C and D.

Implementation comparison (GCC vs NEC4)

Aspect	GCC	NEC4
Initial entitlement	No automatic relief; Clause 13(2), Clause 14.	CE if experienced-contractor test met; cl. 60.1(12).
Administrator	Engineer instructs and assesses; Clause 15, Clause 60(1).	Project Manager decides, instructs, implements; cl. 61.4, cl. 66.1.
Time notice	Delay notice within 28 days; Clause 50(1)(a).	CE notice within eight weeks; cl. 61.3.
Cost notice	Cost notice within 28 days; Clause 64(2).	Quotation within three weeks after instruction; cl. 62.3.
Assessment basis	Retrospective actual cost/variation valuation; Clause 61, Clause 63.	Prospective Defined Cost plus Fee; cl. 63.1.
Mitigation sanction	Proceed with due diligence; Clause 86(1).	Assess as if early warning given; cl. 61.5, cl. 63.7.

5. GCC Lessons for NEC Adoption — GCC routines become NEC management tools.

Hong Kong practitioners already trained to issue Clause 50(1)(a) delay notices and Clause 64(2) cost notices should transfer that discipline to NEC cl. 15.1 and cl. 61.3. The adjustment is that NEC notice is not only defensive claim protection: early warning is given as soon as the obstruction could increase Prices or delay Completion, while CE notice must follow within eight weeks of becoming aware.

The familiar GCC practice of keeping Clause 64(3) contemporary records, preparing Clause 64(4) interim accounts, and supplying Clause 50(3)/Clause 64(6) particulars transfers to NEC cl. 62.2, cl. 62.3, cl. 63.1 and, for Options C, D, E and F, cl. 52.2. The behavioural change is from proving a historical claim to substantiating a quotation: actual Defined Cost to the dividing date, forecast Defined Cost thereafter, Fee, programme effect and risk allowances.

GCC variation routines under Clause 15, Clause 60(1), Clause 61 and Clause 61(1) remain useful when remnant piles require redesign. Under NEC, the equivalent commercial workflow is the compensation-event quotation and PM assessment process under cl. 61.4, cl. 62.2, cl. 62.3, cl. 64.1, cl. 66.1 and cl. 66.2. The adjustment is to price consequences prospectively and collaboratively rather than wait for final measurement or claim evaluation.

GCC EOT analysis under Clause 50(2), Clause 50(3) and programme revision under Clause 50(4) transfers to

NEC Accepted Programme practice under cl. 31.2 and delay assessment under cl. 63.5. In short, GCC claim, record, variation and delay-analysis habits should be used earlier under NEC as risk-management, quotation and mitigation tools.

GCC practice / experience	Transfers to NEC implementation
Clause 50(1)(a) delay notice workflow.	Use cl. 15.1 and cl. 61.3 early; notify before delay crystallises.
Clause 64(3) contemporary records.	Support cl. 63.1 Defined Cost and cl. 52.2 records with auditable data.
Clause 64(4) interim account preparation.	Convert into cl. 62.2 quotation: Prices, Completion Date, programme effects.
Clause 61 variation valuation practice.	Price CE under cl. 63.1 using forecast Defined Cost plus Fee.
Clause 50(2) EOT critical-path analysis.	Maintain cl. 31.2 Accepted Programme for cl. 63.5 Completion Date assessment.

6. HK-Specific Notes — PAH and DEVB notes shape HK practice.

For GCC Mega Project Contracts, the Project Administration Handbook (PAH) Chapter 5, Appendix 5.36 prescribes Special Conditions of Contract Clause A4(2). SCC Clause A4(2) expressly writes mitigation into the contract: “The Contractor shall use and continue to use all reasonable endeavours to avoid or reduce the effects of a cause of delay”. It also imposes a stricter reporting duty than the base GCC response alone, requiring the Contractor to submit interim written particulars at intervals of not more than 28 days until the actual delay is ascertainable.

Under Works Bureau Technical Circular WB1900 Appendix B, the contract administrator’s claim powers are constrained. The Architect/Engineer must report all claims and delays to the Employer and must “refer the principles underlying his assessment of each claim... to enable the Employer to provide his view of the matter before the Architect/Engineer/Surveyor reaches a decision.” This reinforces the Hong Kong public-works practice of careful, documented Engineer assessment rather than informal settlement.

For NEC4 ECC in Hong Kong, the DEVB Practice Notes dated July 2023 / December 2024 provide explicit guidance on unforeseen physical conditions. Section A6.3.1.10 reminds Project Managers that cl. 60.3 enforces the “contra proferentem” rule for inconsistencies in Site Information, so ambiguities are interpreted against the Employer/Client who drafted the document. Sections A6.3.1.32 to A6.3.1.35 clarify that compensation-event assessment inherently includes prolongation costs, including extended use of time-related resources; the Contractor must include the effect of prolongation in the quotation, and the evaluation is made by reference to the period when the effect was felt. Those sections reinforce that NEC assessment is prospective, not postponed retrospectively until the end of the contract.

DEVB Practice Note Section A6.3.1.5 reinforces the difference principle in cl. 60.1(12) and cl. 60.2: only the difference between the physical conditions encountered and those for which it would have been reasonable to have allowed is taken into account. The Project Manager must request documentation on the Contractor’s original risk allowances to assess reasonableness. Section A6.1.1.20 warns that if the Contractor fails to give an early warning, the Project Manager will assess the compensation event as if the warning had been given under cl. 63.7; under Options C and D, the Project Manager may even treat costs incurred due to the lack of an early warning as Disallowed Cost.

Rule-based Validation Summary

Overall verdict: PASSED
Confidence: 1.00
Iterations used: 1
Rule-based track: pass

Metric	Value	Threshold	Status
Citation Accuracy Rate (GCC)	1.00	≥ 0.85	pass
Citation Accuracy Rate (NEC4)	1.00	≥ 0.85	pass
Coverage Rate (GCC)	1.00	≥ 0.85	pass
Coverage Rate (NEC4)	1.00	≥ 0.85	pass
Composite Confidence	1.00	≥ 0.70	pass

GCC anchors covered: Clause 13, Clause 14, Clause 50(1), Clause 50(1)(a), Clause 64

NEC4 anchors covered: cl. 60.1(12), cl. 15.1, cl. 15.2, cl. 63.1, cl. 63.5

LLM Semantic Validation (Track 2)

Track verdict: pass
Severity: minor
Default codes: E1 (hallucinated clause), E2 (egregious content mismatch)
Diagnostic-only — does NOT gate the overall verdict.

No semantic issues raised.

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